

Weight Tables – Joist Girders

The Joist Girders represented in the following Weight Tables are based on the **Steel Joist Institute 45th Edition Standard Specifications - SJI 100-2020**. The following Weight Tables are also based on the use of 50 ksi maximum yield strength steel.

These Weight Tables are intended to be a tool to assist in the preliminary design and estimate for Joist Girders used in floors and roofs. All of the JOIST GIRDER WEIGHT values shown are approximate and intended as a guide for the specifying professional. New Millennium will design for the specific loads of the designation at the required span, and the values for self-weight may vary from the tabulated values – the tabulated values are not design minimums or maximums. It is presumed that the designated kip load includes an allowance for the Joist Girder self-weight, unless noted otherwise on the structural drawings.

There are countless combinations of span, number of panels, kip loads, and Joist Girder depths and the tables do not represent all available combinations. Interpolation can be used for approximate values when needed between the columns and rows of the table.

The Weight Tables apply only to Joist Girders with parallel chords. The Joist Girder top chords are considered as being laterally supported by positive attachment of the supported steel joists to the Joist Girder top chords.

The top (2) rows of figures present the total PANEL POINT LOAD to be applied to the Joist Girder at each joist location for **LRFD** (factored) or **ASD** (unfactored) loading. The JOIST GIRDER WEIGHT values shown in the tables are based on the PANEL POINT LOAD value including an allowance for an assumed Joist Girder self-weight.

Joist Girders weighing up to 149 plf are anticipated to have chord angles of 6 x 6 or smaller and shall have a 7 1/2 inch minimum bearing seat depth.

Joist Girders weighing 150 plf or greater are considered high capacity Joist Girders requiring chord angles of 8 x 8 and shall have a 10 inch minimum bearing seat depth.

Example:

<u>Joist Girder Geometry</u>	<u>Design Loads</u>	
Joist Girder Depth	40"	12 kips ASD Total Load at each joist location
Joist Girder Span	50'-0"	
Number of joist spaces	8	

For this example the Joist Girder designation will be **40G8N12K**

Entering the Weight Tables for a GIRDER SPAN (ft) of 50, JOIST SPACES of 8, a GIRDER DEPTH (in) of 40 and an ASD PANEL POINT LOAD (kips) of 12, the Joist Girder will have the following approximate self-weight/requirements:

JOIST GIRDER WEIGHT = 59 (plf)

Minimum bearing seat depth = 7 1/2"

Standard Weight Tables for Joist Girders

JOIST GIRDER STANDARD WEIGHTS																				
PANEL POINT LOAD (kips)		ASD	4	6	8	10	12	14	16	18	20	24	28	32	36	40	44	48	52	56
		LRFD	6	9	12	15	18	21	24	27	30	36	42	48	54	60	66	72	78	84
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
20	2 spaces @ 10.00'	20	16	19	19	19	19	20	24	24	25	30	37	41	46	50	56	62	70	75
		24	16	19	19	19	19	20	21	21	25	28	32	36	41	42	49	52	53	66
		28	16	19	19	19	19	20	20	21	23	26	28	32	39	40	42	46	48	49
	3 spaces @ 6.67'	20	15	15	19	19	20	23	24	27	31	36	44	48	54	74	75	81	84	89
		24	15	16	16	16	19	20	23	26	27	33	36	45	47	53	56	68	79	82
		28	15	16	16	16	17	20	24	24	26	31	36	44	46	49	53	57	68	80
	4 spaces @ 5.00'	20	15	15	19	21	25	29	33	38	41	50	57	65	71	88	97	100	107	120
		24	15	16	17	20	23	26	29	32	35	44	50	55	62	71	85	90	100	102
		28	16	16	17	19	22	25	28	30	34	39	49	50	59	63	72	86	91	91
	5 spaces @ 4.00'	20	15	17	21	26	31	36	39	48	51	62	71	82	99	99	109	120	141	142
		24	16	16	20	23	26	30	35	39	43	53	60	68	80	91	101	103	110	120
		28	16	16	18	22	27	28	33	37	39	48	55	64	68	77	93	95	107	111
	6 spaces @ 3.33'	20	16	19	25	29	36	41	50	57	58	72	82	99	107	118	138	141		
		24	16	18	22	28	31	37	43	46	53	61	70	85	102	102	111	123	144	147
		28	17	18	22	26	30	33	40	42	47	58	68	76	83	96	109	112	119	130
	8 spaces @ 2.50'	20	19	25	32	41	51	58	65	72	82	99	118	139	142					
		24	17	22	29	36	42	50	54	61	69	86	103	107	128	149	153			
		28	18	22	29	34	40	47	54	61	67	76	88	107	112	124	135	155	166	
22	2 spaces @ 11.00'	20	21	21	21	22	22	23	24	24	25	34	39	43	49	55	62	69	76	78
		24	18	21	21	22	22	22	23	24	24	30	33	41	41	45	51	55	61	73
		28	18	21	21	21	22	22	22	23	24	37	30	33	41	42	46	48	51	58
	3 spaces @ 7.33'	20	15	18	18	19	22	24	26	29	33	42	45	53	68	70	76	84	88	94
		24	15	15	19	19	20	23	24	26	30	35	40	45	48	55	61	74	81	84
		28	15	16	16	16	19	20	23	24	27	32	36	45	47	52	54	59	74	82
	4 spaces @ 5.50'	20	15	16	19	23	26	30	36	39	44	55	62	71	82	95	96	106	119	134
		24	15	15	17	20	25	27	29	34	38	48	52	58	71	79	89	98	101	107
		28	16	16	16	19	22	25	28	32	35	40	49	54	60	72	79	87	90	97
	5 spaces @ 4.40'	20	15	17	24	27	34	38	42	49	55	65	75	96	98	111	126	137		
		24	16	16	20	24	28	33	38	40	48	56	62	73	85	100	101	110	116	133
		28	16	16	18	22	26	30	32	38	41	51	57	65	73	86	92	102	105	111
	6 spaces @ 3.67'	20	16	21	27	33	39	49	56	57	65	79	97	106	118	137				
		24	16	19	23	28	32	39	45	51	58	66	82	98	101	109	120	142	144	
		28	16	18	22	26	30	34	39	44	50	61	70	76	89	102	104	113	127	148
	8 spaces @ 2.75'	20	19	27	36	43	56	64	71	80	96	106	135	138						
		24	18	24	31	38	46	53	60	68	75	101	105	125	145	149				
		28	18	22	28	34	40	47	54	62	69	79	87	106	118	131	152	164		



Standard Weight Tables for Joist Girders

Table of Contents

General Joist Information

Economical Design Guide

Top Chord Ext., K-Series

Joist Substitutes and Outriggers

KCS Joists

Joist Minimum Shear

Standard LRFD Load Tables

Standard ASD Load Tables

Load/Load and SP Designations

Joist Girder Weight Tables

Codes and Standards

JOIST GIRDER STANDARD WEIGHTS

PANEL POINT LOAD (kips)		ASD	4	6	8	10	12	14	16	18	20	24	28	32	36	40	44	48	52	56
		LRFD	6	9	12	15	18	21	24	27	30	36	42	48	54	60	66	72	78	84
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
25	3 spaces @ 8.33'	20	18	18	19	22	26	27	30	37	41	49	59	66	70	76	86	89	97	102
		24	15	18	19	20	22	25	26	28	32	39	43	51	59	67	71	81	84	89
		28	15	15	19	19	20	23	24	27	29	34	39	45	47	55	59	67	81	82
		32	15	16	16	16	20	21	23	24	27	32	36	44	46	52	54	58	74	81
		36	16	16	16	17	17	20	24	24	26	32	36	40	45	48	53	54	68	79
	4 spaces @ 6.25'	20	15	18	20	25	29	35	39	42	49	55	70	78	93	99	109	119	134	135
		24	15	16	19	21	26	29	33	37	40	50	57	64	72	88	97	100	106	120
		28	15	15	17	20	24	25	29	34	37	43	51	58	66	72	89	90	101	102
		32	16	16	17	19	21	25	28	32	35	40	49	54	60	69	79	86	91	96
		36	16	16	17	19	21	26	26	29	34	38	49	50	56	63	73	85	88	92
	5 spaces @ 5.00'	20	15	18	25	31	38	43	51	55	58	73	93	100	109	125	134			
		24	15	17	23	26	32	36	42	47	53	61	75	81	98	102	112	129	140	
		28	16	16	20	24	28	31	37	41	47	56	62	72	79	93	101	106	117	125
		32	16	16	19	23	26	30	33	38	41	51	57	65	73	83	93	102	105	111
		36	16	17	18	22	26	28	31	36	39	48	54	64	69	75	88	96	101	108
	6 spaces @ 4.17'	20	16	24	29	38	45	55	58	69	78	94	104	116	134					
		24	16	20	25	31	37	44	50	56	64	75	97	99	107	118	138			
		28	16	18	23	28	32	38	44	51	55	67	73	87	101	104	120	134	143	145
		32	16	18	22	26	30	34	39	44	50	61	69	77	89	102	105	113	127	148
		36	16	18	24	25	30	36	39	43	49	58	67	74	84	98	108	116	117	129
	8 spaces @ 3.12'	20	21	29	39	48	58	70	78	94	99	115	134							
		24	19	26	33	41	50	57	65	75	81	99	118	138						
		28	18	23	30	38	44	53	60	67	75	86	103	116	127	147				
		32	18	24	28	34	39	47	54	65	71	78	87	105	117	129	152	154		
		36	18	22	29	34	40	46	52	61	63	76	87	101	114	121	136	148	166	167
	10 spaces @ 2.50'	20	26	38	49	63	78	94	100	115	134									
		24	23	33	42	54	65	75	89	99	104	130								
		28	21	30	38	48	56	64	74	84	101	109	134	147						
		32	21	28	36	43	52	62	69	76	87	107	118	130	153					
		36	22	28	37	44	52	64	71	77	85	100	116	130	151	157				
28	3 spaces @ 9.33'	24	18	18	19	22	24	27	29	36	39	43	53	62	70	71	78	85	89	98
		28	18	18	19	20	22	25	26	28	31	39	43	46	55	61	66	76	83	86
		32	15	18	19	19	21	23	24	27	28	34	39	45	48	53	58	66	80	81
	4 spaces @ 7.00'	24	15	16	20	24	27	32	38	40	48	55	62	71	82	95	104	106	120	135
		28	15	15	18	21	25	28	32	36	39	49	56	64	71	79	96	97	106	107
		32	15	15	17	20	23	25	29	33	37	43	50	58	62	70	85	90	99	102
	5 spaces @ 5.60'	24	15	18	24	29	34	39	46	52	58	66	78	96	102	111	126	136		
		28	15	17	21	26	30	35	39	46	50	61	68	77	90	99	107	114	130	142
		32	16	17	20	24	27	32	37	41	44	56	62	70	80	93	102	107	112	119
	6 spaces @ 4.67'	24	16	21	28	35	41	49	55	63	70	79	96	106	134	137				
		28	15	20	24	30	36	42	50	54	58	71	82	99	107	118	138	142		
		32	16	19	23	28	32	37	43	49	53	64	74	84	101	102	111	123	144	146
	7 spaces @ 4.00'	24	18	24	32	41	49	56	64	74	79	96	110	135						
		28	17	22	27	35	43	51	57	62	69	82	99	108	129	140				
		32	16	21	27	31	38	44	52	55	63	74	85	102	108	123	143	146		
	8 spaces @ 3.50'	24	20	28	37	48	55	64	74	79	95	105	134							
		28	18	25	32	39	50	58	65	72	81	99	108	129	141					
		32	17	24	29	38	43	53	60	64	70	86	103	113	127	147	149			
	10 spaces @ 2.80'	24	24	36	46	57	70	79	96	102	117	137								
		28	23	30	41	50	60	69	82	99	100	120	141							
		32	21	30	38	46	55	66	71	80	93	109	126	147						



Discover the easiest way to specify Steel Joists and Joist Girders:

www.newmill.com/digital-tools

Standard Weight Tables for Joist Girders

JOIST GIRDER STANDARD WEIGHTS																				
PANEL POINT LOAD (kips)		ASD	4	6	8	10	12	14	16	18	20	24	28	32	36	40	44	48	52	56
		LRFD	6	9	12	15	18	21	24	27	30	36	42	48	54	60	66	72	78	84
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
30	3 spaces @ 10.00'	24	18	18	21	24	27	31	35	38	40	48	58	66	71	80	92	98	117	119
		28	18	18	19	22	25	27	30	35	37	42	49	56	63	70	79	82	93	99
		32	18	18	19	20	22	26	28	31	32	39	46	51	57	64	71	73	83	84
		36	16	19	19	19	21	23	26	28	31	35	39	46	52	57	64	65	73	75
	4 spaces @ 7.50'	24	16	18	23	29	33	37	42	49	53	64	76	85	101	104	126	127	149	150
		28	15	16	21	25	30	33	37	42	45	53	61	73	81	86	103	104	126	128
		32	15	16	18	22	26	30	34	37	43	51	55	62	70	77	87	103	105	116
		36	16	16	17	22	24	27	31	34	36	46	52	59	64	74	78	88	91	105
	5 spaces @ 6.00'	24	15	19	25	30	37	43	51	55	58	73	86	96	109	125	134			
		28	15	17	23	27	32	37	44	47	53	61	75	88	97	102	112	128	138	
		32	16	17	21	24	29	35	39	43	48	56	63	77	90	100	101	107	117	133
		36	16	17	20	24	27	31	36	40	43	51	60	70	80	86	94	103	110	118
	6 spaces @ 5.00'	24	16	24	29	37	45	52	58	66	73	94	104	116	134					
		28	16	20	27	32	38	44	50	57	65	75	97	99	107	137	140			
		32	16	19	24	29	34	40	45	51	58	65	82	98	100	109	121	142	144	
		36	16	18	23	26	31	37	41	46	52	61	70	84	101	102	111	123	126	148
	8 spaces @ 3.75'	24	21	32	40	51	63	73	83	99	111	124	146							
		28	20	30	37	44	53	61	73	80	86	114	126	149						
		32	18	26	34	42	49	55	63	71	79	104	117	130	154	161				
		36	17	23	32	39	46	54	61	69	76	89	108	121	134	154	169			
	10 spaces @ 3.00'	24	25	38	51	66	78	99	111	123	134									
		28	24	36	47	57	69	80	94	113	116	138								
		32	22	31	39	52	58	74	82	95	105	129	142							
		36	22	30	39	48	54	68	79	84	91	119	132	151						
32	3 spaces @ 10.67'	24	18	19	21	26	27	34	38	40	42	54	61	70	75	84	88	102	102	113
		28	16	17	18	24	26	28	31	34	37	43	55	60	69	70	76	85	89	93
		32	17	17	18	21	25	26	28	32	34	39	44	54	61	62	67	77	80	86
		36	15	17	19	20	23	25	26	28	30	38	40	45	51	53	58	67	81	77
	4 spaces @ 8.00'	24	18	19	23	26	32	37	40	47	55	61	72	86	94	103	114	133	134	
		28	15	18	20	24	28	32	37	40	45	55	62	70	78	94	96	105	121	135
		32	15	15	20	22	25	29	32	36	39	49	56	64	71	83	82	97	102	107
		36	15	16	17	21	24	26	30	34	36	43	50	58	65	70	85	90	99	102
	5 spaces @ 6.40'	24	15	20	27	33	39	44	51	57	65	77	93	100	123	133				
		28	15	18	24	28	34	39	46	52	58	66	74	96	101	110	126	137		
		32	15	17	22	26	32	35	41	46	53	61	68	77	90	99	105	114	130	142
		36	16	17	21	24	27	33	37	42	47	56	62	70	79	93	102	106	117	120
	6 spaces @ 5.33'	24	17	24	31	39	47	55	61	69	76	94	103	133	134					
		28	16	21	27	35	40	48	55	60	67	79	96	105	117	137				
		32	16	20	25	30	36	42	50	54	58	71	82	99	103	118	139	142		
		36	16	19	24	28	34	38	44	49	55	66	73	84	101	102	111	123	144	146
	8 spaces @ 4.00'	24	22	32	40	54	61	72	86	93	103	133								
		28	19	27	35	45	55	63	70	80	95	105	134	137						
		32	18	25	32	39	50	58	65	71	81	99	109	120	141					
		36	18	24	31	38	43	53	59	67	71	86	103	113	127	147				



Standard Weight Tables for Joist Girders

Table of Contents

General Joist Information

Economical Design Guide

Top Chord Ext., K-Series

Joist Substitutes and Outriggers

KCS Joists

Joist Minimum Shear

Standard LRFD Load Tables

Standard ASD Load Tables

Load/Load and SP Designations

Joist Girder Weight Tables

Codes and Standards

JOIST GIRDER STANDARD WEIGHTS

PANEL POINT LOAD (kips)		ASD	4	6	8	10	12	14	16	18	20	24	28	32	36	40	44	48	52	56
		LRFD	6	9	12	15	18	21	24	27	30	36	42	48	54	60	66	72	78	84
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
35	4 spaces @ 8.75'	28	16	19	23	27	31	36	41	46	52	60	74	79	94	100	111	117	137	138
		32	15	18	21	24	28	33	37	39	45	53	60	73	80	92	100	106	112	127
		36	15	16	20	23	27	30	33	37	41	51	55	62	74	83	94	97	107	113
		40	15	16	17	21	26	27	30	37	38	46	52	61	64	75	90	95	96	108
	5 spaces @ 7.00'	28	15	20	26	32	37	43	52	57	59	73	86	100	109	126	136			
		32	15	18	24	29	34	37	45	50	53	66	75	88	100	102	112	128	138	
		36	16	17	23	27	29	35	40	46	48	62	68	77	90	100	104	115	131	133
		40	16	17	22	25	27	33	37	43	47	56	63	70	80	95	102	107	115	125
	6 spaces @ 5.83'	28	17	24	30	37	44	52	58	65	73	93	103	115	134					
		32	16	21	27	33	38	46	53	57	65	79	96	100	117	139	140			
		36	16	20	25	31	36	41	48	54	58	70	81	99	102	113	121	142	144	
		40	16	20	24	28	34	38	44	49	55	64	77	84	101	104	115	123	145	146
	7 spaces @ 5.00'	28	19	27	34	43	52	59	66	74	86	101	115	135						
		32	17	24	30	39	47	53	61	67	75	97	103	118	137					
		36	17	23	28	35	42	48	55	62	69	82	99	105	120	141	144			
		40	17	22	27	32	39	44	50	55	63	73	86	102	107	118	133	147		
	8 spaces @ 4.38'	28	21	30	39	48	59	69	78	94	98	115	136							
		32	20	27	36	42	53	61	69	79	88	101	118	138						
		36	19	26	32	39	48	55	62	71	77	99	109	121	141					
		40	18	24	30	37	44	54	60	65	73	86	102	113	127	147	149			
38	4 spaces @ 9.50'	32	16	19	21	26	31	34	39	43	48	58	67	74	87	100	101	111	127	138
		36	15	17	21	24	28	33	35	39	44	53	60	74	75	93	97	106	112	123
		40	15	16	20	23	27	30	34	37	41	51	55	62	74	83	94	98	107	109
		44	16	16	20	22	26	28	30	35	38	46	52	58	65	75	90	95	95	108
	5 spaces @ 7.60'	32	15	20	25	31	36	42	46	52	59	70	86	96	101	111	126	137		
		36	16	20	24	28	33	38	45	47	53	64	74	89	98	103	112	129	138	
		40	16	20	23	26	31	35	40	46	48	59	70	78	91	101	105	113	117	134
		44	17	20	22	25	30	33	39	41	48	56	63	75	80	93	102	107	111	118
	6 spaces @ 6.33'	32	17	24	30	35	41	49	55	62	70	86	98	105	125	136				
		36	16	21	27	33	39	47	50	57	61	75	89	100	107	118	141	142		
		40	16	21	25	31	36	40	48	55	59	71	82	99	102	109	121	143	142	
		44	17	20	24	29	33	38	44	49	55	64	77	84	102	104	115	123	145	147
	8 spaces @ 4.75'	32	20	29	38	47	56	64	74	86	95	105	135							
		36	19	28	35	42	50	57	65	76	81	101	113	138	140					
		40	19	26	32	40	48	55	62	67	78	100	103	121	142	144				
		44	20	24	30	39	47	51	57	64	71	86	102	113	127	147	149			
40	4 spaces @ 10.00'	32	17	20	23	29	37	40	47	50	56	64	73	86	103	114	126	128	149	151
		36	17	19	22	29	31	37	40	44	51	57	65	74	87	103	104	125	127	128
		40	17	18	22	25	29	33	37	40	47	52	62	73	77	87	96	104	117	127
		44	16	17	20	24	29	31	36	38	41	49	59	66	74	78	84	96	106	106
	5 spaces @ 8.00'	48	17	17	20	24	25	30	32	37	39	48	53	59	67	78	78	85	99	106
		32	15	21	26	32	38	43	52	55	62	73	86	101	109	124	134			
		36	16	20	24	30	34	39	45	53	55	66	74	88	102	102	112	128	138	
		40	16	20	24	27	32	37	41	46	51	62	68	77	90	100	105	115	130	142
	6 spaces @ 6.67'	44	17	20	23	29	32	37	41	49	50	58	70	82	84	99	116	118	130	141
		48	17	20	23	26	31	34	40	41	50	57	68	75	85	95	100	119	120	132
		32	16	24	30	38	44	52	58	65	72	93	100	115	133					
		36	17	22	27	34	39	47	53	60	67	79	97	102	117	137	141			
	7 spaces @ 5.71'	40	16	21	26	30	36	43	48	54	62	71	82	99	103	114	130	142		
		44	17	21	24	28	36	40	47	51	55	66	78	91	102	107	116	134	142	146
		48	17	21	24	31	36	42	46	53	57	69	79	86	100	109	132	133	135	164
		32	18	26	33	43	52	58	66	74	86	101	115	135						
	8 spaces @ 5.00'	36	17	24	31	39	47	53	61	67	75	97	103	117	136					
		40	17	24	29	35	43	49	55	62	69	82	99	105	119	140				
		44	20	22	28	33	39	48	55	59	64	78	92	102	111	122	143			
		48	20	23	28	36	41	48	54	61	66	80	86	108	122	134	136	164	167	
	10 spaces @ 4.00'	32	21	29	38	48	58	67	78	94	96	115	135							
		36	19	27	36	46	53	60	68	80	88	102	118	137						
		40	19	25	34	39	49	58	65	72	82	99	109	120	141					
		44	21	27	33	39	47	56	63	70	75	93	103	120	136	147				
		48	20	25	32	42	47	55	62	69	80	90	104	122	136	155	170			
		32	29	39	51	64	79	92	112	123	125	149								
		36	25	36	47	60	69	81	94	103	125	150								
		40	24	36	45	56	66	75	82	96	115	129	152							
		44	23	32	41	51	60	71	82	84	99	119	143	161						
		48	23	32	41	52	58	68	76	85	94	121	134	152						



Discover the easiest way to specify Steel Joists and Joist Girders:

www.newmill.com/digital-tools

Standard Weight Tables for Joist Girders

JOIST GIRDER STANDARD WEIGHTS																				
PANEL POINT LOAD (kips)		ASD	4	6	8	10	12	14	16	18	20	24	28	32	36	40	44	48	52	56
		LRFD	6	9	12	15	18	21	24	27	30	36	42	48	54	60	66	72	78	84
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
42	4 spaces @ 10.50'	32	16	21	25	29	34	38	43	49	53	67	74	86	99	101	112	125	134	138
		36	16	19	22	26	32	35	39	44	47	58	67	73	87	95	101	112	118	129
		40	16	19	21	24	28	34	36	41	45	53	61	73	76	93	97	97	113	122
		44	16	19	20	23	27	31	34	38	42	51	55	62	74	84	94	97	108	109
	5 spaces @ 8.40'	48	16	19	21	24	26	29	32	36	39	47	54	62	65	75	90	95	97	108
		32	16	22	28	35	41	45	52	57	66	74	88	100	110	125				
		36	15	21	25	31	36	42	46	52	59	70	85	96	102	111	126	137		
		40	16	21	24	28	33	39	44	51	54	64	74	89	98	103	113	129	130	
	6 spaces @ 7.00'	44	16	20	24	27	31	37	40	46	52	59	69	78	91	101	105	113	126	134
		48	17	20	23	27	30	35	39	42	48	57	63	75	81	95	102	107	115	118
		32	18	25	32	39	45	55	61	69	77	93	103	124	135					
		36	17	23	30	35	41	49	56	60	67	79	96	105	117	137				
	7 spaces @ 6.00'	40	17	21	26	33	39	46	54	57	61	75	89	100	108	119	141	142		
		44	16	21	24	31	35	41	48	54	59	71	81	100	102	109	121	143	142	
		48	20	20	25	29	33	39	44	49	56	64	77	85	102	104	115	124	145	147
		32	20	28	36	45	52	65	72	85	93	102	125							
	8 spaces @ 5.25'	36	19	26	34	40	49	56	67	74	79	98	110	127	138					
		40	18	24	31	38	46	54	61	68	75	90	101	113	129	142				
		44	20	23	29	35	41	49	55	63	70	78	100	106	116	132	145			
		48	18	23	28	34	39	44	50	56	64	73	92	102	108	118	136	149		
	10 spaces @ 4.20'	32	22	32	40	51	62	72	78	94	100	124	135							
		36	20	27	38	46	56	64	74	79	96	105	126	138						
		40	20	26	35	42	51	57	65	76	81	101	113	138	141					
		44	20	25	32	39	49	55	63	70	78	99	107	121	142	147				
45	4 spaces @ 11.25'	48	21	26	32	41	48	56	63	67	74	93	103	112	128	148				
		32	27	38	52	62	77	94	101	114	134									
		36	25	36	46	60	70	86	97	102	112	140								
		40	24	34	45	54	64	75	89	99	104	129								
	5 spaces @ 9.00'	44	23	31	41	52	61	70	79	91	100	114	143							
		48	23	30	39	49	56	66	72	80	93	107	125	146						
		36	18	21	25	28	33	38	42	46	52	62	72	79	95	100	112	117	128	138
		40	19	21	22	27	31	35	39	44	47	55	64	75	87	95	101	112	113	128
	6 spaces @ 7.50'	44	19	21	22	24	29	33	37	39	45	53	61	74	76	89	95	102	108	114
		48	18	21	22	24	28	31	34	38	40	51	55	63	75	83	94	95	107	109
		52	18	22	23	24	27	29	33	37	39	47	52	60	66	76	91	95	96	109
		36	16	22	27	33	38	44	52	55	63	74	86	101	109	125	136			
	7 spaces @ 6.43'	40	16	21	25	30	36	42	45	53	56	68	75	88	102	111	122	128		
		44	16	21	24	29	34	38	44	46	54	65	74	85	90	103	110	123	130	142
		48	20	21	24	27	32	36	41	45	52	59	67	75	91	95	106	112	118	134
		52	20	21	24	27	30	35	39	42	48	57	64	75	81	94	98	107	117	119
	8 spaces @ 5.62'	36	19	24	31	38	45	52	58	66	74	93	100	115	134					
		40	19	23	28	34	40	47	53	60	67	79	97	103	117	137	140			
		44	19	21	27	32	38	46	50	54	62	76	90	100	107	118	139	142		
		48	20	21	26	30	36	42	48	55	59	69	78	92	102	110	122	143	143	
	9 spaces @ 5.00'	52	20	21	25	29	34	39	44	50	56	64	77	85	102	102	116	124	136	148
		36	20	27	35	44	52	58	66	74	86	101	115	135						
		40	20	26	33	40	47	54	61	67	75	97	105	127	138					
		44	20	24	30	39	46	54	61	62	69	90	100	113	129	143				
	10 spaces @ 4.50'	48	20	23	29	36	41	49	55	63	70	79	92	107	117	133	145			
		52	18	23	28	34	39	45	50	56	65	73	93	102	109	118	136	149		
		36	21	30	38	48	58	67	78	94	98	114	135							
		40	20	28	36	46	53	61	68	80	89	105	118	137						
		44	20	27	34	41	51	58	66	73	81	99	109	130	141					
		48	21	26	32	39	47	55	63	68	74	92	104	116	142	146				
		52	22	28	33	42	48	54	59	67	71	94	102	112	127	148				
		36	24	34	45	55	66	74	88	98	104	135								
		40	22	31	39	49	61	69	80	89	100	113	138							
		44	23	31	39	48	58	66	76	89	99	108	132							
		48	23	29	37	47	55	63	70	79	91	106	117	133						
		52	23	28	36	46	55	60	70	73	84	102	112	135	148					
		36	26	38	49	60	73	86	98	105	116	137								
		40	25	35	47	60	66	76	90	102	112	140								
		44	24	33	46	54	64	72	89	99	104	130	142							
		48	24	31	40	49	62	71	78	91	100	114	134							
		52	23	31	39	50	56	67	72	80	93	107	123	147						



Standard Weight Tables for Joist Girders

Table of Contents

General Joist Information

Economical Design Guide

Top Chord Ext., K-Series

Joist Substitutes and Outriggers

KCS Joists

Joist Minimum Shear

Standard LRFD Load Tables

Standard ASD Load Tables

Load/Load and SP Designations

Joist Girder Weight Tables

Codes and Standards

JOIST GIRDER STANDARD WEIGHTS

PANEL POINT LOAD (kips)		ASD	4	6	8	10	12	14	16	18	20	24	28	32	36	40	44	48	52	56
		LRFD	6	9	12	15	18	21	24	27	30	36	42	48	54	60	66	72	78	84
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
48	5 spaces @ 9.60'	36	19	26	31	37	45	52	59	66	71	87	111	113	135	136				
		40	19	23	29	35	41	46	52	59	68	77	92	112	114	136	138			
		44	19	22	27	32	37	44	48	54	61	69	80	93	113	116	126	139	150	
		48	19	21	25	30	36	40	48	48	55	69	78	90	96	115	116	128	140	142
		52	20	21	25	29	33	39	42	50	54	62	71	82	92	99	117	118	130	141
	6 spaces @ 8.00'	36	20	21	24	29	33	38	40	46	50	59	71	79	85	100	100	119	120	133
		40	20	28	35	42	51	62	70	78	83	100	122	134	147					
		44	19	25	33	39	47	56	64	71	79	93	112	124	137	148				
		48	19	24	31	36	45	50	57	65	73	81	102	115	127	138	151			
		52	20	23	27	32	38	46	51	59	60	75	83	97	107	130	131	144	162	
	8 spaces @ 6.00'	36	20	22	27	31	37	42	48	54	61	69	80	91	107	120	132	134	153	165
		40	30	36	45	56	64	78	91	100	122	134								
		44	28	33	42	51	59	70	80	92	101	124	148							
		48	27	32	39	49	55	65	74	82	95	114	127	150						
		52	26	30	37	47	53	60	68	76	84	105	129	131	154					
	9 spaces @ 5.33'	40	26	30	36	44	51	59	65	71	80	99	119	132	146	164				
		44	25	28	36	43	49	57	63	69	78	90	109	123	136	155				
		48	36	44	55	70	79	91	99	121	122	146								
		52	40	34	42	52	63	74	88	93	101	113	136							
		56	33	39	50	59	69	83	91	94	103	126	150							
	12 spaces @ 4.00'	48	33	37	46	56	66	76	85	94	97	118	130							
		52	31	36	46	54	63	72	80	95	101	108	132	152						
		56	31	35	44	53	62	69	80	89	98	103	123	137	165					
		36	35	52	71	84	100	123	135	148										
		40	34	48	65	76	93	113	125	137	149									
50	5 spaces @ 10.00'	44	31	44	57	73	82	102	115	126	139									
		48	30	41	53	67	76	88	104	117	130	153								
		52	30	39	52	61	76	84	97	107	131	144								
		56	27	38	49	61	70	81	91	108	122	135	165							
		40	18	23	30	38	44	47	56	60	68	79	93	113	124	136	138			
	6 spaces @ 8.33'	44	17	22	29	34	40	46	51	56	61	76	89	94	113	126	137	139		
		48	19	22	28	31	38	42	48	55	61	69	78	94	96	115	127	139	141	
		52	20	22	25	31	35	40	45	49	55	62	74	82	96	116	117	129	141	142
		56	20	22	25	30	32	40	43	50	51	63	71	83	92	99	117	119	131	142
		60	20	20	24	30	33	36	42	46	51	58	65	76	86	96	101	120	121	133
	8 spaces @ 6.25'	40	20	28	34	42	48	56	64	71	80	100	112	124	147					
		44	19	24	31	38	47	50	57	65	73	85	102	124	127	149				
		48	19	23	30	37	40	49	57	65	67	82	95	115	127	129	151			
		52	20	23	30	36	40	46	52	59	67	75	84	105	117	129	131	153	162	
		56	20	23	26	33	39	42	51	54	60	72	84	98	107	120	132	144	163	164
	10 spaces @ 5.00'	60	21	23	27	33	38	43	49	53	61	70	80	87	102	110	123	134	154	165
		40	22	31	39	51	59	67	78	86	96	110	135							
		44	21	29	37	47	53	61	70	80	96	103	118	139						
		48	21	27	35	42	51	58	69	76	81	99	114	130	142					
		52	21	25	33	40	49	55	63	70	78	99	107	121	141					
	12 spaces @ 4.17'	56	24	29	36	42	47	56	64	68	78	94	108	118	137	148				
		60	24	27	35	40	47	55	61	69	74	83	103	110	123	139	149			
		40	24	34	44	55	66	74	86	96	104	134								
		44	23	32	40	53	61	69	80	88	98	113	138							
		48	24	32	42	52	58	69	77	90	99	111	133							



Discover the easiest way to specify Steel Joists and Joist Girders:

www.newmill.com/digital-tools

Standard Weight Tables for Joist Girders

JOIST GIRDER STANDARD WEIGHTS																				
PANEL POINT LOAD (kips)		ASD	4.0	5.0	6.0	7.0	8.0	9.0	10.0	11.0	12.0	13.0	14.0	16.0	18.0	20.0	22.0	24.0	26.0	28.0
		LRFD	6.0	7.5	9.0	10.5	12.0	13.5	15.0	16.5	18.0	19.5	21.0	24.0	27.0	30.0	33.0	36.0	39.0	42.0
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
55	5 spaces @ 11.00'	44	21	21	24	25	29	32	35	38	41	43	47	53	59	63	71	82	83	86
		48	21	21	23	24	28	30	32	35	38	41	43	49	56	60	64	71	73	83
		52	20	22	23	25	27	29	32	33	36	39	42	44	52	57	65	66	74	74
		56	20	21	24	24	26	28	31	33	36	37	39	44	51	53	58	66	66	74
	6 spaces @ 9.17'	60	23	24	24	24	27	31	33	35	38	38	45	47	52	60	61	67	68	68
		66	24	24	24	25	26	28	33	34	37	37	42	47	48	55	56	62	62	69
		44	19	22	26	29	33	36	38	43	45	51	52	59	66	75	86	86	98	101
		48	20	22	24	28	31	33	36	40	44	46	50	56	64	68	75	87	89	98
	7 spaces @ 7.86'	52	20	22	24	26	29	33	35	37	41	59	59	66	74	86	93	99	109	110
		56	18	21	24	25	28	31	35	36	39	42	47	52	55	63	70	71	78	91
		60	20	21	24	25	29	30	33	35	38	39	43	48	55	60	64	71	75	80
		66	19	20	22	24	28	30	31	33	36	39	40	47	50	56	62	65	73	73
	9 spaces @ 6.11'	44	21	24	28	33	36	39	44	50	53	59	59	70	75	87	97	102	111	120
		48	21	24	27	31	34	38	43	45	51	54	56	65	72	76	89	98	103	110
		52	21	23	26	29	33	36	39	44	46	52	55	62	69	74	86	91	100	105
		56	20	22	25	28	31	35	38	40	46	48	53	55	64	70	79	87	92	101
	11 spaces @ 5.00'	60	21	22	24	27	30	33	36	39	41	47	49	56	64	68	72	81	93	94
		66	22	22	24	26	30	32	36	37	40	43	48	52	58	65	70	74	83	84
		44	24	29	34	39	46	52	55	60	67	74	74	87	98	105	116	135	137	
		48	24	28	32	38	40	47	53	57	61	68	69	81	97	103	107	118	129	139
60	5 spaces @ 12.00'	52	25	30	33	39	43	47	52	57	65	65	73	77	90	104	105	114	125	133
		56	24	29	32	38	43	46	51	53	59	66	67	75	87	92	105	107	117	128
		60	24	27	32	36	40	45	47	52	56	60	67	71	80	93	95	108	109	118
		66	24	27	31	35	39	42	46	49	54	58	61	71	78	83	91	97	111	113
	6 spaces @ 10.00'	44	30	36	43	49	55	63	67	74	87	88	97	106	126	137				
		48	28	33	39	45	54	61	65	69	76	87	89	103	112	128	139			
		52	27	34	37	44	52	55	62	66	73	77	88	99	105	115	131	142		
		56	27	33	39	42	48	54	60	64	68	77	80	93	102	107	118	134	146	
	8 spaces @ 7.50'	60	26	31	37	40	47	49	58	64	67	72	77	82	95	108	110	121	137	148
		66	26	31	36	39	45	50	54	60	65	68	74	82	97	98	113	117	126	141
		48	21	23	27	29	33	35	39	43	44	49	51	57	63	69	76	87	89	94
		52	21	22	27	28	31	33	36	40	44	45	47	52	60	65	69	77	85	90
	10 spaces @ 6.00'	56	22	23	24	28	30	31	34	36	41	44	45	52	59	63	69	74	78	87
		60	22	23	24	28	29	32	34	35	40	42	45	49	53	60	66	70	75	80
		66	24	24	24	26	30	30	33	35	36	38	42	47	51	56	61	67	72	73
		72	25	25	25	25	27	30	31	35	36	37	39	45	48	56	56	63	69	70
	12 spaces @ 5.00'	48	20	24	29	32	36	38	41	47	49	56	60	67	72	80	93	93	112	113
		52	20	23	28	30	33	37	39	46	48	50	57	62	69	78	80	94	94	113
		56	19	24	25	30	33	38	39	42	48	49	51	58	66	69	79	83	95	96
		60	19	23	24	29	32	34	39	40	43	49	50	57	63	70	75	83	83	96
	15 spaces @ 4.00'	66	19	23	24	27	32	32	34	40	42	44	50	52	61	65	69	77	84	85
		72	22	22	24	27	28	33	34	36	41	43	44	52	54	63	68	71	75	87
		48	24	29	34	39	43	49	56	57	64	72	72	80	93	112	123	125	136	148
		52	23	29	31	37	40	48	50	57	58	66	72	81	94	103	114	125	127	139
	12 spaces @ 5.00'	56	23	26	31	36	38	44	49	51	58	60	66	75	83	96	104	116	127	129
		60	23	26	32	33	39	42	47	50	53	59	61	69	77	85	98	106	118	129
		66	28	30	33	34	41	43	46	48	53	57	62	70	78	82	90	100	108	120
		72	29	30	31	34	36	41	46	47	52	58	59	66	73	80	90	92	104	110
	10 spaces @ 6.00'	48	26	32	37	44	49	55	60	67	74	79	87	97	105	118	137	138		
		52	28	34	38	44	50	56	64	65	71	75	88	97	103	113	130	138		
		56	27	33	37	43	46	51	58	66	65	72	76	90	104	105	123	131	143	
		60	25	31	37	39	45	51	57	60	66	70	73	86	93	104	111	126	134	
	12 spaces @ 5.00'	66	27	32	37	42	49	51	56	62	65	72	74	85	95	102	120	122	134	145
		72	26	32	33	38	42	47	50	55	59	66	69	74	83	96	98	111	111	121
		48	33	39	46	53	59	68	75	86	87	97	102	111	135					
		52	31	37	45	51	57	65	69	76	88	89	98	104	118	139				
	15 spaces @ 4.00'	56	29	36	41	48	55	62	66	72	77	89	91	104	113	129	140			
		60	30	35	39	47	54	56	64	73	74	79	91	102	106	116	133	145		
		66	32	35	41	48	53	61	62	70	77	80	87	100	110	122	134	147	164	
		72	29	33	38	42	50	52	60	61	69	72	77	86	100	110	114	127	142	151
	15 spaces @ 4.00'	48	40	49	64	72	80	93	102	113	124	126	136							
		52	39	48	57	66	74	81	94	103	114	126	127	150						
		56	38	46	53	67	71	80	83	96	104	116	127	140	153					
		60	38	42	51	60	68	76	83	89	98	106	118	132	144					
	15 spaces @ 4.00'	66	35	41	49	55	62	70	81	87	87	98	103	110	123	136	153	167		
		72	35	44	46	55	64	66	77	85	90	93	106	125	139	142	160	171		

